# SleepSentry: A Novel Sleep Apnea Detection System in the Internet of Things (IoT)

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2025 IEEE International Conference on Consumer Electronics (ICCE) | 979-8-3315-2116-5/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICCE63647.2025.10930207

***Abstract*—Sleep apnea (SA) is a sleep disorder that affects 5 to 10 percent of the world's population and about 39 million adults in the United States. It disrupts sleep either by partial stoppage or complete blockage of breathing and leads to serious health conditions if left untreated. Current diagnostic solutions for sleep apnea are costly and time-consuming. In this work, a convolutional neural network (CNN) based fast and automated sleep apnea detection system has been presented which provides comprehensive solutions for patient healthcare. The use of CNN after filtering the test data makes the system highly optimized and fast. The integration of Internet of Things (IoT) technology enables data to be securely pushed to servers for further analysis and user notifications. The publicly accessible PhysioNet dataset is used for validation, and the proposed method improves upon existing solutions, making it a strong contender for application in wearable medical devices.**

***Index Terms*—Internet of Things (IoT), Convolutional Neural Network(CNN), Sleep Apnea (SA), Electrocardiogram(ECG), Blood Oxygen Saturation(Spo2)**

## I. Introduction

Sleep apnea is a chronic ailment characterized by repeated episodes of breathing pauses during sleep. Sleep apnea can be categorized into three cases: complex sleep apnea, central sleep apnea (CSA), and obstructive sleep apnea(OSA) [1]. OSA is the most common condition in which the patient is not able to breathe either completely or intermittently. The early signs and symptoms of this condition may include gasping for air, chronic snoring, night sweats, morning headaches, impaired memory, and sleep disruptions. This issue can further complicate and lead to other serious ailments if not treated early, such as heart attack and diabetes [2].

Sleep apnea mostly goes undetected in as much as 75 percent of the cases. Therefore, detecting it early and treating it is crucial. However, the current methods for detecting sleep apnea involve tests that can only be conducted in labs, commonly referred to as polysomnography (PSG), where patients are monitored using multiple sensors to track various physiological signals, such as brain waves, electrocardiogram (ECG), oxygen saturation (SpO2), heart rate, breathing, eye movements, and muscle activity. Such tests can be costly and also time-consuming for patients [3]. Taking all of this into account, the primary objective of the research topic is to develop a reliable, scalable, and accessible system for the continuous monitoring and fast detection of sleep apnea, which can also alert emergency services in life-threatening situations. Through the use of machine learning techniques, this work aims to train on and learn from datasets of existing patients, utilizing convolutional neural networks (CNN) [4] [1] [5] and long short-term memory (LSTM) methods to detect sleep apnea using physiological signals such as ECG and SpO2. After detecting sleep apnea, Sleepsentry will transmit the data to the ThingSpeak servers for further analysis and logging. If the patient is found to be experiencing a severe episode, the relevant emergency services will be notified. SleepSentry contributes to consumer electronics by delivering a low-power, real-time sleep apnea detection solution using IoT and cloud computing. It enhances smart healthcare devices with accurate, continuous monitoring, improving health management for users. Fig. 1 depicts sleep apnea condition.

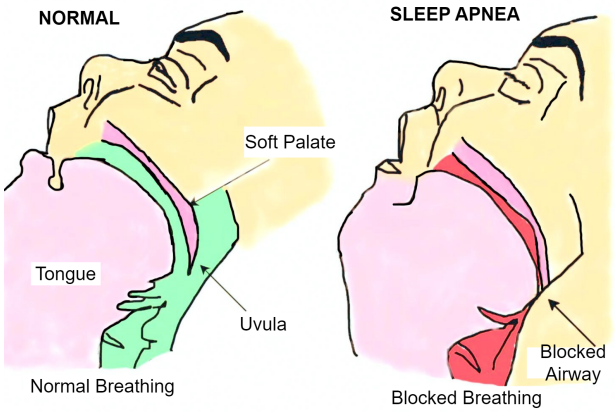



Fig. 1. Sleep Apnea Condition

The remainder of the paper is organized as follows. Section II of the paper highlights the key innovations of this work. In Section III, the paper reviews existing research on sleep apnea detection system, contrasting the limitations of current methods with the advantages of the SleepSentry system. Section IV discusses the proposed approach, including data collection, data cleanup for pre-processing, and the machine learning algorithm used. Section V presents the numerical results and analysis of the proposed system. Finally, this paper is concluded in Section VI, where the key findings

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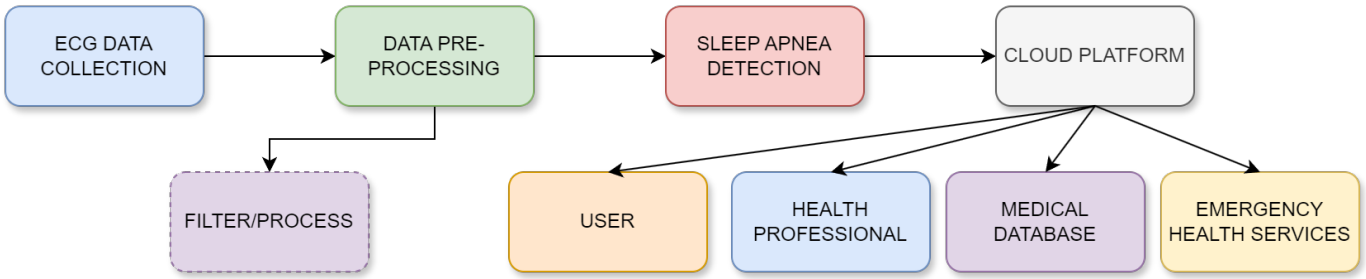



Fig. 2. Architecture for the Proposed SleepSentry

are summarized, the significance of the SleepSentry system is reiterated, and potential implications for future research are discussed

## II. Novel contribution

1) Existing approaches used stationary data to detect sleep apnea. This work implements real-time detection by introducing a fixed-length time window, enabling continuous and real-time analysis of ECG and SpO2 data. SleepSentry provides a comprehensive end-to-end solution for fast detection, data processing, learning, and emergency notification to health care services.
2) Proposed system bridges the gap between detection and treatment by integrating an automated alert system that notifies emergency healthcare services upon detecting critical sleep apnea events. This helps us to ensure timely medical intervention, which can potentially save countless lives.
3) Sleep sentry leverages the power of Convolutional Neural Network (CNN) to analyze ECG and Spo2 data, demonstrating their effectiveness in detecting sleep apnea with higher accuracy and speed compared to traditional machine learning algorithms. Sleep sentry digitizes the data into smaller chunks in JSON format and creates segments that can be processed by the system to train, which eventually lets us train the model faster.

## III. Related Research

Various methods have been proposed for the SA detection. A composite deep convolutional neural network model [1] has been proposed for obstructive sleep apnea detection that utilizes single-lead ECG signals to capture the complex dynamics of the biomedical ECG signals. Another SA detection scheme [3] employed wearable devices, specifically photoplethysmography, for rapid SA detection, highlighting the use of various sensor modalities to enhance detection accuracy. In the Edge-IoT based SA detection approach [6], ECG signals are analyzed in both the cloud node and the edge node to validate the effectiveness of edge computing platform. Edge node reported a fast SA detection while maintaining high detection accuracy compared to cloud based system, making it a viable candidate for faster SA detection. The effectiveness of CNN derived from photoplethysmography (PPG) signals is investigated in this research [7], contributing to the classification of sleep stages, which is relevant for understanding sleep disorders such as sleep apnea. The integration of multimodal and multi-resolution data [4] using convolutional neural networks enables effective SA detection and holds potential for use in wearable healthcare applications.

The existing methods focus on the detection aspect of sleep apnea and those algorithms analyzed signals to find patterns for sleep apnea conditions. However, there is a lack of comprehensive solutions that combine continuous monitoring, early detection, and integration with emergency healthcare services using IoT. This work developed a comprehensive, reliable, time-sensitive, and cost-effective method of continuous measurement and early sleep apnea detection using machine learning algorithms and Internet of Things (IoT).

## IV. Proposed Novel Sleep Apnea Detection Algorithm

SleepSentry consists of three main components: data preprocessing, convolutional neural networks (CNNs), and integration with ThingSpeak. The architecture of the proposed system is shown in Fig. 2.

### A. Data Pre-Processing

After the data has been collected, the information from the .dat files or digitized ECG signals is denoised and segmented into JSON format. The segment size is set to one minute, as ".apn data" is available in one-minute intervals, providing 6000 sample points for the ECG data (100 samples per second). Later on, the system will be calibrated for 30-second intervals since QRS data is also available. The Python script reads the minute-by-minute ECG data from the .dat files and the apnea detection data from the .apn files, then generates a set of JSON files to be used for training the machine learning models. The fields currently being used are Patient ID, Patient Category, Patient Age, Sleep Time, 60 seconds of samples for ECG, and Apnea Markers. The data is being JSONized so that the models can be fed easily.

### B. Convolutional neural network (CNN)

The system aims to utilize 1D CNN [8] [4] and Long Short-Term Memory (LSTM) methods to classify the data, followed by training the model using the learning datasets. This model intends to use ECG data in the beginning labeled with an

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apnea flag and later incorporate Spo2 data into the model to increase efficiency. Preliminary calculations indicate that using a multi-modal approach, an accuracy of over 90% can be achieved in detecting sleep apnea, outperforming existing machine learning algorithms. Furthermore, due to the multi-modal nature of the system, different subtypes of sleep apnea can be differentiated. After the model has been properly trained, the goal is to feed it with fresh data from new patients. The structure of the CNN is depicted in Fig. 3.

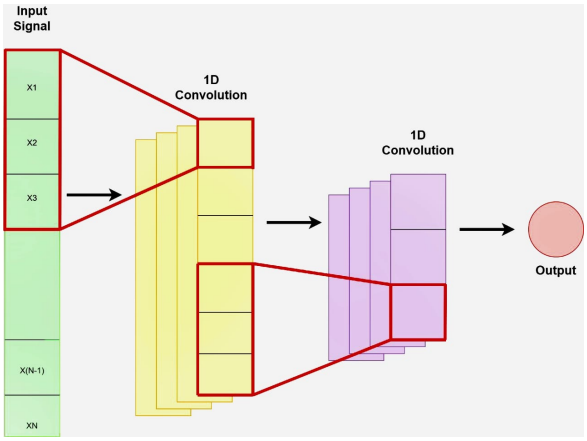



Fig. 3. Convolutional Neural Network

As the model is already equipped to detect sleep apnea, once any abnormalities are detected, it will connect to the ThingSpeak servers to push the real-time data. On the portal, the real-time data will be visible along with options to analyze and study it. If any critical conditions are identified, the server can send emergency notifications to the relevant emergency services, the patient, and the family doctor. Therefore, the findings and results indicate that the SleepSentry provides a more reliable and comprehensive solution for detecting early cases of sleep apnea, helping doctors intervene sooner and improve patient outcomes.

### C. Integration of ThingSpeak

As soon as the patient's data is received on the local device, SleepSentry pushes the data to the ThingSpeak servers, where it is split into two streams: one consisting of all the data samples for future analysis, and the other containing 60-second samples when sleep apnea is detected beyond a certain threshold, along with the apnea marker. On the ThingSpeak server, the data can be further plotted and analyzed by a physician for a more granular study. The data can also be used by Sleep Sentry to improve its training. The overview of ThingSpeak integration is demonstrated in Fig. 4.

If it is found that the patient is going into prolonged sleep apnea and breathing has stopped completely, SleepSentry triggers a notification to the emergency health services, family doctor, and also the emergency contacts of the patient.

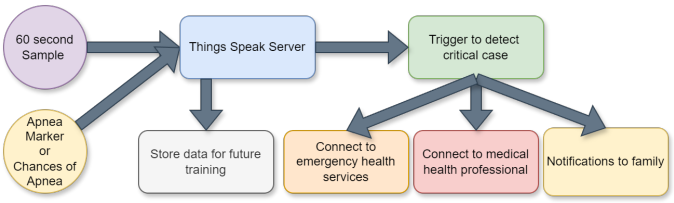



Fig. 4. Things Speak Server End

## V. Implementation and Validation of the Proposed SleepSentry

Data is obtained from the Apnea-ECG Database created by George Moody and Roger Mark (MIT Laboratory for Computational Physiology, Cambridge, USA) [9] on PhysioNet. The data consists of 70 samples from patients divided into two sets of 35 records as shown in Table I. The first set of records is the learning set, which consists of 3 categories of patients: a, b, and c. Category a has the highest sleep apnea occurrences, with at least 100 minutes of detected sleep apnea, b contains between 5 and 99 minutes of detected sleep apnea, and finally, c with 0 to 4 minutes of detected apnea. The length of the records varies from 7 hours to 10 hours each. The ECG data, which were the .dat files and contained the continuous digitized ECG signal, was used. Each ECG data is 16 bits per sample, with the least significant byte first in each pair. The data was collected at a rate of 100 samples per second, nominally 200 A/D units per millivolt. The .hea files are the header files, which include names and formats of the specified signal/data files. The .apn files contain the data for the apnea incidents marked per minute by human experts, whereas the .qrs contain the machine-generated binary annotation signals for sleep apnea. Finally, the other set of 35 samples, which are the test data sets that have the apnea marker white out, can be used to test the efficiency of the model created.

Data cleanup process is performed before feeding the data into the model. The ECG data is shown in Fig. **??**. Fig. 6 shows the ECG samples in 60 sec window. The experiments were conducted using a dataset that contained ECG data collected from patients with varying degrees of sleep apnea. The data set was further divided into training and testing sets. The training set of each patient was then segmented and filtered for noise. The data was segmented into 60-second samples. The data set contains 100 samples in each second, therefore creating 6000 samples of ECG data per 60 seconds. Along with the ECG data, the APN file marked with an apnea marker was read and matched with that of the 60-second ECG data.

Initially, the model was trained using the CPU, but CUDA was used to increase the training speed, which increased the training performance by 40 percent. System Specifications Used:
CPU : Ryzen 5 3600

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TABLE I
TRAINING DATA

| Patient data used for training Sleep Sentry | | | | | | |
|---|---|---|---|---|---|---|
| Record | Sample Length | Non-Apnea (Mins) | Apnea (Mins) | AHI | Age | Weight (kg) |
| a01 | 490 | 20 | 470 | 69.6 | 51 | 102 |
| a02 | 529 | 109 | 420 | 69.5 | 38 | 120 |
| a03 | 520 | 274 | 246 | 39.1 | 54 | 80 |
| a04 | 493 | 40 | 453 | 77.4 | 52 | 121 |
| a05 | 455 | 179 | 276 | 41 | 58 | 78 |
| a06 | 511 | 305 | 206 | 24.7 | 63 | 104 |
| a07 | 512 | 190 | 322 | 63 | 44 | 105 |
| a08 | 502 | 313 | 189 | 42 | 51 | 88 |
| a09 | 496 | 115 | 381 | 31.7 | 52 | 82 |
| a10 | 518 | 418 | 100 | 21 | 58 | 78 |
| a11 | 467 | 245 | 222 | 14 | 58 | 103 |
| a12 | 578 | 44 | 534 | 80.2 | 52 | 121 |
| a13 | 496 | 252 | 244 | 42 | 51 | 88 |
| a14 | 510 | 127 | 383 | 54.7 | 51 | 102 |
| a15 | 511 | 143 | 368 | 52 | 60 | 113 |
| a16 | 483 | 163 | 320 | 41 | 44 | 105 |
| a17 | 486 | 328 | 158 | 33 | 40 | 96 |
| a18 | 490 | 52 | 438 | 82.4 | 52 | 82 |
| a19 | 503 | 298 | 205 | 34 | 55 | 90 |
| a20 | 511 | 196 | 315 | 41 | 58 | 78 |
| b01 | 488 | 469 | 19 | 0.24 | 44 | 63 |
| b02 | 518 | 425 | 93 | 19 | 53 | 85 |
| b03 | 442 | 369 | 73 | 24 | 53 | 85 |
| b04 | 430 | 420 | 10 | 0.7 | 42 | 64 |
| b05 | 434 | 377 | 57 | 5 | 52 | 135 |
| c01 | 485 | 485 | 0 | 0 | 31 | 74 |
| c02 | 503 | 502 | 1 | 0 | 37 | 83 |
| c03 | 455 | 455 | 0 | 0 | 39 | 65 |
| c04 | 483 | 483 | 0 | 0 | 41 | 65 |
| c05 | 467 | 464 | 3 | 0 | 28 | 57 |
| c06 | 469 | 468 | 1 | 0.25 | 28 | 65 |
| c07 | 454 | 450 | 4 | 0 | 30 | 56 |
| c08 | 535 | 535 | 0 | 0 | 42 | 64 |
| c09 | 469 | 467 | 2 | 0 | 37 | 83 |
| c10 | 432 | 431 | 1 | 0 | 27 | 72 |

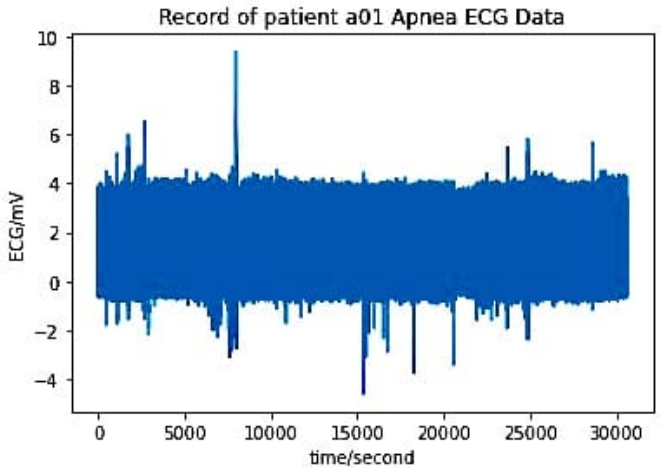



Fig. 5. Patient a01 ECG data

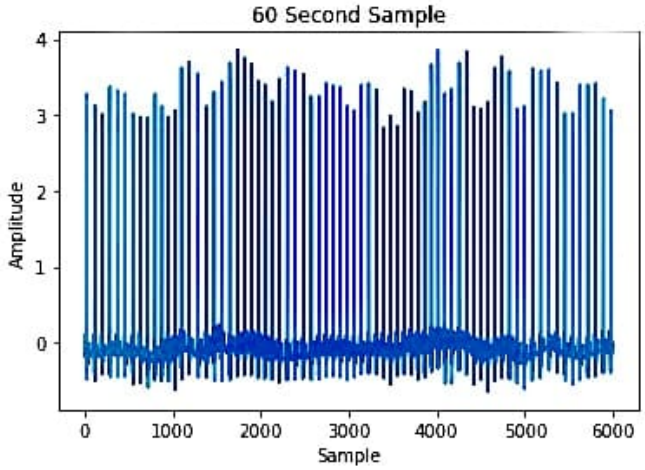



Fig. 6. 6000 ECG Samples in 60 sec Window (Patient a01)

GPU : NVIDIA 1070 TI
CUDA CORES : 2432
Core Speed : 1607 MHz

The dataset was prepared using ECG data from Sleep Apnea patients with varying degrees of the condition. The ECG data and corresponding apnea labels were loaded and converted into PyTorch tensors. The data set was split into training and validation sets using the "torch.utilts.data.random_split" function. Next, training and validation data loaders are created using 'DataLoader' objects to enable batch-wise processing. The CNN model used has two convolution layers ("conv1" and "conv2") followed by max-pooling layers to extract features from the input ECG signals. The output of the convolutional layers is flattened and passed through two fully connected layers (fc1 and fc2) for classification. The CNN model was trained on the training dataset using a cross-entropy loss function and the Adam optimizer. The training was performed over multiple epochs, with model parameters updated using backpropagation. After training, the trained model was evaluated using the validation dataset to assess its performance. Evaluation metrics were calculated to quantify the model's ability to correctly classify apnea and non-apnea instances. The simulations were conducted using Python with the use of libraries such as TensorFlow, JSON, WFDB, NUMPY.

After the initial training with varied datasets as depicted in Fig. 7, SleepSentry was able to detect sleep apnea with an accuracy of 80.1%, a sensitivity of 79.28%, and a specificity of 78.23% using the 1D CNN for worst-case scenarios (Fig. 8 ). This included data from patients with over 80% sleep apnea samples in 8 to 10 hours of sleep data, as well as samples from patients with about 1% to 2% sleep apnea samples during the same duration. Sleep Sentry was trained using data from patients labeled a and b, indicating a high prevalence of sleep apnea. This training resulted in an estimation accuracy of approximately 89.70%. Proposed system is compared to existing methods, as shown in Fig. 9, which demonstrates that

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SleepSentry outperforms the existing methods [3] [10] [6] [1].

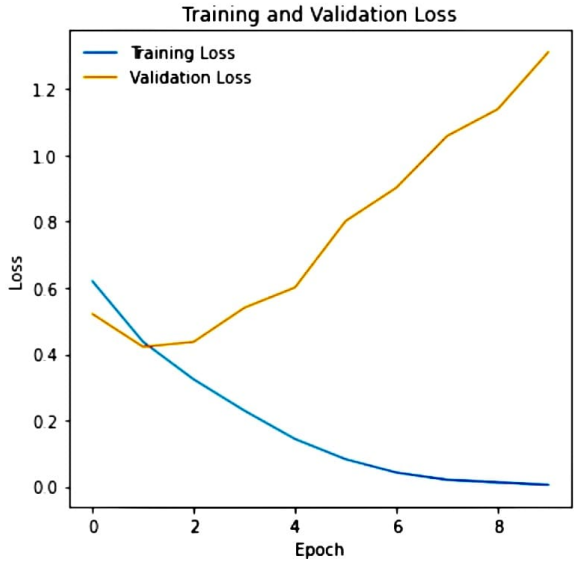



Fig. 7. Training and Validation Loss

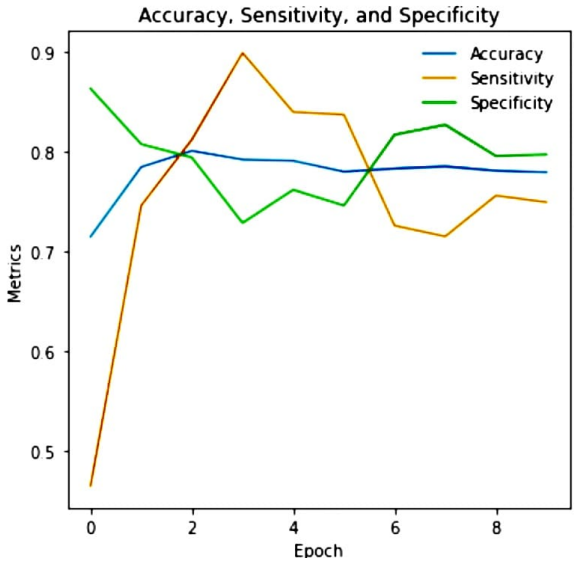



Fig. 8. Accuracy, Specificity, and Sensitivity

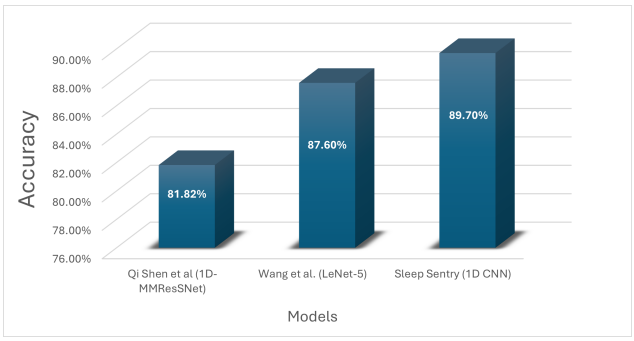



Fig. 9. Comparison to Existing Works

## VI. Conclusions

The proposed SleepSentry aims to provide a comprehensive, real-time, and fast detection solution for Sleep apnea diagnosis. Experimental results show that Sleepsentry provides a more efficient and user-friendly solution for detecting and managing SA. Future research will focus on incorporating security measures to protect the system from external attacks. Given the vulnerability of sensitive health information to external attacks, implementing encryption protocols for data transmission to the cloud will be explored. Future research will also include creating a more compact version of the proposed system by utilizing state-of-the-art integrated circuit technologies designed for wearable biomedical applications.

## References


[1] Y. Zhou, Y. He, and K. Kang, "OSA-CCNN: Obstructive Sleep Apnea Detection Based on a Composite Deep Convolution Neural Network Model using Single-Lead ECG signal," in *2022 IEEE International Conference on Bioinformatics and Biomedicine*, pp. 1840–1845, 2022.

[2] C. . Y. Dental., "Sleep Apnea Treatment in Irvine Orange, CA." https://www.chuyangdental.com/services/sleep-apnea/, 2023. [Online; accessed 29-April-2024].

[3] Q. Shen, X. Yang, L. Zou, K. Wei, C. Wang, and G. Liu, "Multitask Residual Shrinkage Convolutional Neural Network for Sleep Apnea Detection Based on Wearable Bracelet Photoplethysmography," *IEEE Internet of Things Journal*, vol. 9, no. 24, pp. 25207–25222, 2022.

[4] A. John, K. K. Nundy, B. Cardiff, and D. John, "Multimodal Multiresolution Data Fusion Using Convolutional Neural Networks for IoT Wearable Sensing," *IEEE Transactions on Biomedical Circuits and Systems*, vol. 15, no. 6, pp. 1161–1173, 2021.

[5] Y. Zhang, A. Banta, Y. Fu, M. M. John, A. Post, M. Razavi, J. Cavallaro, B. Aazhang, and Y. Lin, "RT-RCG: Neural Network and Accelerator Search Towards Effective and Real-time ECG Reconstruction from Intracardiac Electrograms," *ACM Journal on Emerging Technologies in Computing Systems*, vol. 18, mar 2022.

[6] M. A. Sayeed and P. Carr, "An Accurate and Low-power Sleep Apnea Detector in the Edge-IoT Platform, year=2023," in *2023 IEEE International Conference on Artificial Intelligence, Blockchain, and Internet of Things*, pp. 1–4.

[7] A. Habib, M. A. Motin, T. Penzel, M. Palaniswami, J. Yearwood, and C. Karmakar, "Performance of a Convolutional Neural Network Derived From PPG Signal in Classifying Sleep Stages," *IEEE Transactions on Biomedical Engineering*, vol. 70, no. 6, pp. 1717–1728, 2023.

[8] M. Shenfield, Alex Howarth, "A Novel Deep Learning Model for the Detection and Identification of Rolling Element-Bearing Faults." https://www.researchgate.net/figure/, 2020. [Online; accessed 29-April-2024].

[9] A. L. Goldberger, L. A. Amaral, L. Glass, J. M. Hausdorff, P. C. Ivanov, R. G. Mark, J. E. Mietus, G. B. Moody, C. K. Peng, and H. E. Stanley, "PhysioBank, PhysioToolkit, and PhysioNet: components of a new research resource for complex physiologic signals," *Circulation*, vol. 101, pp. E215–20, June 2000.

[10] T. Wang, C. Lu, G. Shen, and F. Hong, "Sleep apnea detection from a single-lead ECG signal with automatic feature-extraction through a modified LeNet-5 convolutional neural network," *the Journal of Life Environmental Sciences (PeerJ)*, vol. 7, p. e7731, 09 2019.


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